American Inventors: Rocket Scientist Robert Goddard

Today rocket launches and space missions are common. But in the early 1900s, space travel seemed like a dream.

One of the most influential people in the field of rocket science was American Robert Goddard.

The American space agency NASA describes Goddard as “the father of modern rocket propulsion.”

Robert Goddard once said that "the dream of yesterday is the hope of today and the reality of tomorrow." His scientific work gave hope to many dreams about space travel. He turned some of those dreams into reality.

More than one hundred years ago, Goddard carried out studies and tests of rocket engines.

Robert H. Goddard is shown June 8, 1938 at Roswell, New Mexico, with one of the rockets he developed from 1926 to 1941. (AP Photo)

He developed and flew many rockets that got their power from solid fuels: chemicals that formed a hard substance. In 1925, he made and tested the first rocket engine using a soft chemical fuel. The next year, he successfully launched the world's first liquid-fuel rocket.

Many historians consider liquid-fuel rocket flight to be as important as the first airplane flight by the American brothers Orville and Wilbur Wright.

Goddard's work proved that machines could travel outside of Earth's atmosphere and into space.

During his early research, he received money and support from the U.S. Smithsonian Institution. The Smithsonian published several reports about his efforts.

Dr. Robert H. Goddard at the blackboard at Clark University, Worcester, MA, in 1924. (AP photo)

One publication, called “A Method of Reaching Extreme Altitudes,” wrote about his search for ways to send weather recording instruments higher than balloons could fly. It described how he developed the mathematical theories for rocket flight.

In that report, Goddard also suggested the possibility of a rocket someday reaching the moon. At the time, there was a big dispute in the press about this claim. Many people thought he was foolish for suggesting something that seemed so impossible.

Many of Goddard's ideas are still used in rocket development. So, in a way, every rocket that flies today could be considered a Goddard rocket.

I'm Bryan Lynn.

Shirley Griffith reported on this story for VOA Learning English. Bryan Lynn adapted it.

Quiz - American Inventors: Rocket Scientist Robert Goddard

Vocabulary

mission - n. ภารกิจ; การเดินทางในอวกาศที่มีเป้าหมาย

propulsion - n. แรงขับเคลื่อน; การผลักดันไปข้างหน้า

Reading Comprehension Quiz

1. What was the general perception of space travel in the early 1900s?

A) It was a common occurrence.

B) It was considered a distant dream.

C) It was already a reality.

D) It was primarily a military objective.

2. Who is recognized by NASA as "the father of modern rocket propulsion"?

A) Orville Wright

B) Wilbur Wright

C) Robert Goddard

D) Bryan Lynn

3. According to Robert Goddard, what is the progression from a 'dream of yesterday'?

A) It becomes a forgotten memory and then a myth.

B) It becomes the hope of today and the reality of tomorrow.

C) It leads to immediate scientific breakthroughs.

D) It causes widespread public doubt.

4. When did Robert Goddard begin his significant studies and tests of rocket engines?

A) In the early 1900s

B) More than one hundred years ago

C) During the 1920s

D) In 1938

5. Which type of fuel did Robert Goddard initially use for many of the rockets he developed and flew?

A) Liquid fuels

B) Soft chemical fuels

C) Solid fuels

D) Gaseous fuels

6. What was Goddard's groundbreaking achievement in the year 1926?

A) He developed the first rocket engine using a soft chemical fuel.

B) He made the first successful launch of a liquid-fuel rocket.

C) He developed rockets powered by solid fuels.

D) He published a report on reaching extreme altitudes.

7. How do many historians view the importance of liquid-fuel rocket flight?

A) As a minor scientific advancement.

B) As comparable to the first airplane flight.

C) As less significant than solid-fuel rockets.

D) As merely a dream that became a hope.

8. What fundamental concept did Goddard's work definitively prove regarding space travel?

A) That rockets could only use solid fuels.

B) That the moon was an impossible target.

C) That machines could travel beyond Earth's atmosphere.

D) That space missions were common in the early 1900s.

9. Which institution provided early financial support and published reports about Goddard's efforts?

A) NASA

B) Clark University

C) The Smithsonian Institution

D) VOA Learning English

10. What was one of the key purposes of Goddard's 'A Method of Reaching Extreme Altitudes' publication?

A) To advocate for moon travel.

B) To describe how to send weather instruments higher than balloons.

C) To prove the feasibility of solid-fuel rockets.

D) To dispute claims about impossible space travel.

11. Besides the practical application mentioned, what other scientific contribution was described in 'A Method of Reaching Extreme Altitudes'?

A) The development of soft chemical fuels.

B) The successful launch of a liquid-fuel rocket.

C) The mathematical theories for rocket flight.

D) The historical comparison to the Wright brothers.

12. In his report, what ambitious possibility did Goddard suggest that caused a 'big dispute'?

A) Developing a new type of solid fuel.

B) Launching weather recording instruments.

C) A rocket someday reaching the moon.

D) Collaborating with the Smithsonian Institution.

13. How did many people in the press react to Goddard's suggestion of a rocket reaching the moon?

A) They supported his visionary ideas.

B) They considered him foolish and the idea impossible.

C) They immediately began funding his research.

D) They requested further scientific proof.

14. What is stated about the relevance of Goddard's ideas in current rocket development?

A) They have been largely replaced by newer theories.

B) They are still used in today's rocket development.

C) They are considered outdated and historical curiosities.

D) They primarily apply to solid-fuel rockets.

15. According to the article, why could every rocket that flies today be considered a 'Goddard rocket'?

A) He personally built all the early prototypes.

B) His specific designs are universally copied.

C) His fundamental ideas are still incorporated.

D) He was the first to launch any rocket.